

Assignment 3: Data Exploration

Josh Salzberg

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
# for this assignment, we're using tidyverse, lubridate and here
library(tidyverse)
library(lubridate)
library(here)

# we are also using 2 datasets: ECOTOX neonicotinoid dataset & Niwot Ridge NEON dataset
ecotoxdf <- read.csv(here('./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
                     stringsAsFactors = T)
neondf <- read.csv(here('./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
                   stringsAsFactors = T)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might be interested in seeing at what level of accumulations are the chemicals in insecticides effective, how the chemicals travel through food chains, or capturing a more complete view of how the chemicals enter the environment beyond the site of application.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might be interested in identifying optimal component admixtures for rich decomposition.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. An elevated and ground-level pair of traps are deployed for every 400 square meter plot areas, resulting in 1-4 trap pairs per plot. 2. The frequency at which traps are checked is dependent on the predominant tree type at the sampling location, with deciduous sites receiving biweekly sampling, while evergreen sites are checked every 1-2 months 3. Trap placement is either ordered or random depending on the vegetation.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
# dim() returns the overall dimensions of a dataframe
dim(ecotoxdf)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#sorting the summary count of factors in the Effect column of ecotoxdf
sort(summary(ecotoxdf$Effect),decreasing = T)
```

```
##      Population      Mortality      Behavior Feeding behavior
##           1803           1493           360             255
##      Reproduction      Development      Avoidance      Genetics
```

##	197	136	102	82
##	Enzyme(s)	Growth	Morphology	Immunological
##	62	38	22	16
##	Accumulation	Intoxication	Biochemistry	Cell(s)
##	12	12	11	9
##	Physiology	Histology	Hormone(s)	
##	7	5	1	

Answer: The 5 most common effects studied are population, mortality, behavior, feeding behavior, and reproduction. These are vital measurements to understand how an insecticide impacts insect populations. Population, mortality and reproduction are often taken together as a trio to understand demographic changes. Behavior and feeding behavior are important to capture how insecticides disrupt the natural function of the insects.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
# sorting the summary count of species common names
sort(summary(ecotoxdf$Species.Common.Name), decreasing = TRUE)
```

##	(Other)	Honey Bee
##	670	667
##	Parasitic Wasp	Buff Tailed Bumblebee
##	285	183
##	Carniolan Honey Bee	Bumble Bee
##	152	140
##	Italian Honeybee	Japanese Beetle
##	113	94
##	Asian Lady Beetle	Euonymus Scale
##	76	75
##	Wireworm	European Dark Bee
##	69	66
##	Minute Pirate Bug	Asian Citrus Psyllid
##	62	60
##	Parastic Wasp	Colorado Potato Beetle
##	58	57
##	Parasitoid Wasp	Erythrina Gall Wasp
##	51	49
##	Beetle Order	Snout Beetle Family, Weevil
##	47	47
##	Sevenspotted Lady Beetle	True Bug Order
##	46	45
##	Buff-tailed Bumblebee	Aphid Family
##	39	38
##	Cabbage Looper	Sweetpotato Whitefly
##	38	37
##	Braconid Wasp	Cotton Aphid
##	33	33
##	Predatory Mite	Ladybird Beetle Family
##	33	30
##	Parasitoid	Scarab Beetle

##	30	29
##	Spring Tiphia	Thrip Order
##	29	29
##	Ground Beetle Family	Rove Beetle Family
##	27	27
##	Tobacco Aphid	Chalcid Wasp
##	27	25
##	Convergent Lady Beetle	Stingless Bee
##	25	25
##	Spider/Mite Class	Tobacco Flea Beetle
##	24	24
##	Citrus Leafminer	Ladybird Beetle
##	23	23
##	Mason Bee	Mosquito
##	22	22
##	Argentine Ant	Beetle
##	21	21
##	Flatheaded Appletree Borer	Horned Oak Gall Wasp
##	20	20
##	Leaf Beetle Family	Potato Leafhopper
##	20	20
##	Tooth-necked Fungus Beetle	Codling Moth
##	20	19
##	Black-spotted Lady Beetle	Calico Scale
##	18	18
##	Fairyfly Parasitoid	Lady Beetle
##	18	18
##	Minute Parasitic Wasps	Mirid Bug
##	18	18
##	Mulberry Pyralid	Silkworm
##	18	18
##	Vedalia Beetle	Araneoid Spider Order
##	18	17
##	Bee Order	Egg Parasitoid
##	17	17
##	Insect Class	Moth And Butterfly Order
##	17	17
##	Oystershell Scale Parasitoid	Hemlock Woolly Adelgid Lady Beetle
##	17	16
##	Hemlock Woolly Adelgid	Mite
##	16	16
##	Onion Thrip	Western Flower Thrips
##	16	15
##	Corn Earworm	Green Peach Aphid
##	14	14
##	House Fly	Ox Beetle
##	14	14
##	Red Scale Parasite	Spined Soldier Bug
##	14	14
##	Armoured Scale Family	Diamondback Moth
##	13	13
##	Eulophid Wasp	Monarch Butterfly
##	13	13
##	Predatory Bug	Yellow Fever Mosquito

##		13		13
##	Braconid Parasitoid		Common Thrip	
##		12		12
##	Eastern Subterranean Termite		Jassid	
##		12		12
##	Mite Order		Pea Aphid	
##		12		12
##	Pond Wolf Spider		Spotless Ladybird Beetle	
##		12		11
##	Glasshouse Potato Wasp		Lacewing	
##		10		10
##	Southern House Mosquito		Two Spotted Lady Beetle	
##		10		10
##	Ant Family		Apple Maggot	
##		9		9

Answer: the six most common species listed are p"Other",] Honey Bee, Parastic Wasp, Buff Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee and Italian Honeybee. Ignoring "Other", we can easily identify these are all pollinators, who play a very important role in producing healthy agricultural outputs, by fertilizing the flowering plants to produce fruit.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
levels(ecotoxdf$Conc.1..Author.)
```

```
## [1] "<0.0004" "<0.025" "<0.088" "<0.5" "<1.5" "<10/"
## [7] "<2.5/" "<4.00" "<5.00" "<8.79" ">=13.66" ">0.01"
## [13] ">0.1" ">0.220" ">0.5" ">100.0" ">1000.00" ">20"
## [19] ">2500.00" ">5.0" ">6300" ">81" ">98" ">98.43"
## [25] "~10" "~30/" "~40/" "~41" "0.000006" "0.0000073"
## [31] "0.000047" "0.000048" "0.00005" "0.000069/" "0.00007" "0.000074"
## [37] "0.00008" "0.0000814" "0.0001" "0.000101" "0.00017" "0.0002"
## [43] "0.0002/" "0.00027/" "0.00039" "0.0004/" "0.0005" "0.0005/"
## [49] "0.00053" "0.0006" "0.00071" "0.00077" "0.00084/" "0.0009"
## [55] "0.001" "0.0011" "0.0013" "0.0014" "0.0015" "0.002"
## [61] "0.0022/" "0.0027" "0.003" "0.003/" "0.0032" "0.00322"
## [67] "0.0035/" "0.00355/" "0.00368" "0.0037/" "0.0039" "0.004"
## [73] "0.005" "0.005/" "0.0056" "0.0063" "0.007" "0.007/"
## [79] "0.00739" "0.0076" "0.008" "0.009" "0.0095" "0.01"
## [85] "0.01/" "0.010725" "0.011" "0.0112" "0.0113" "0.012"
## [91] "0.012/" "0.0125" "0.013" "0.013/" "0.014" "0.015"
## [97] "0.016" "0.017" "0.0179" "0.018" "0.0192/" "0.02"
## [103] "0.02/" "0.0206/" "0.021" "0.0218" "0.022" "0.023"
## [109] "0.0231" "0.024" "0.025" "0.025/" "0.026" "0.026/"
## [115] "0.027" "0.028" "0.0298" "0.0299" "0.03" "0.031"
## [121] "0.032" "0.033" "0.035" "0.037" "0.0373" "0.0375"
## [127] "0.04" "0.04/" "0.0400/" "0.0428" "0.0429/" "0.0439"
## [133] "0.047" "0.047/" "0.048/" "0.05" "0.05/" "0.051"
## [139] "0.053" "0.053/" "0.056" "0.056/" "0.06" "0.06/"
## [145] "0.0600/" "0.061" "0.061/" "0.0626" "0.063" "0.064"
## [151] "0.0641" "0.0671" "0.06825/" "0.07" "0.07/" "0.071"
## [157] "0.075" "0.077" "0.078" "0.08" "0.08/" "0.080/"
```

##	[163]	"0.081"	"0.0812/"	"0.084/"	"0.085"	"0.088"	"0.09"
##	[169]	"0.09/"	"0.091"	"0.1"	"0.1/"	"0.102"	"0.1039"
##	[175]	"0.105"	"0.112"	"0.112/"	"0.113"	"0.114/"	"0.116/"
##	[181]	"0.118"	"0.12"	"0.123"	"0.124"	"0.125"	"0.125/"
##	[187]	"0.126/"	"0.13"	"0.13/"	"0.1332"	"0.134"	"0.137"
##	[193]	"0.138"	"0.14"	"0.14/"	"0.141"	"0.145"	"0.15"
##	[199]	"0.15/"	"0.150/"	"0.1500/"	"0.154"	"0.154/"	"0.155"
##	[205]	"0.16"	"0.16/"	"0.161"	"0.17"	"0.17/"	"0.18"
##	[211]	"0.18/"	"0.2"	"0.2/"	"0.202"	"0.21"	"0.21/"
##	[217]	"0.2175"	"0.225"	"0.225/"	"0.227"	"0.23"	"0.233"
##	[223]	"0.24"	"0.24/"	"0.246"	"0.25"	"0.25/"	"0.26"
##	[229]	"0.26/"	"0.267"	"0.28"	"0.28/"	"0.29"	"0.29/"
##	[235]	"0.291"	"0.295"	"0.297"	"0.3"	"0.3/"	"0.30/"
##	[241]	"0.302"	"0.31"	"0.32"	"0.33/"	"0.336"	"0.34"
##	[247]	"0.35"	"0.36"	"0.37"	"0.37/"	"0.375"	"0.38"
##	[253]	"0.39/"	"0.397"	"0.4"	"0.4/"	"0.40/"	"0.41"
##	[259]	"0.42"	"0.428"	"0.43"	"0.43/"	"0.44"	"0.4483"
##	[265]	"0.45"	"0.45/"	"0.46"	"0.46/"	"0.47"	"0.470/"
##	[271]	"0.48"	"0.481"	"0.49"	"0.49/"	"0.491"	"0.493"
##	[277]	"0.494"	"0.5"	"0.5/"	"0.50/"	"0.51"	"0.52"
##	[283]	"0.52/"	"0.53"	"0.536"	"0.558"	"0.56"	"0.56/"
##	[289]	"0.567/"	"0.57"	"0.58"	"0.580/"	"0.594"	"0.6"
##	[295]	"0.6/"	"0.609"	"0.61"	"0.625"	"0.63/"	"0.64"
##	[301]	"0.647"	"0.692"	"0.699/"	"0.7"	"0.71"	"0.72"
##	[307]	"0.72/"	"0.73"	"0.74/"	"0.745"	"0.75"	"0.75/"
##	[313]	"0.76/"	"0.79"	"0.8/"	"0.83"	"0.84"	"0.87"
##	[319]	"0.88"	"0.9"	"0.91"	"0.93"	"0.93/"	"0.946"
##	[325]	"0.95"	"0.9535"	"0.96"	"0.96/"	"0.98"	"1"
##	[331]	"1.0/"	"1.00/"	"1.01"	"1.04"	"1.051"	"1.07"
##	[337]	"1.08"	"1.09/"	"1.092"	"1.1"	"1.11"	"1.12"
##	[343]	"1.12/"	"1.16"	"1.2"	"1.20/"	"1.21"	"1.24"
##	[349]	"1.25"	"1.25/"	"1.28"	"1.28/"	"1.29"	"1.3"
##	[355]	"1.3/"	"1.34/"	"1.35"	"1.37"	"1.38"	"1.4"
##	[361]	"1.43/"	"1.44"	"1.45"	"1.48/"	"1.5"	"1.5/"
##	[367]	"1.51"	"1.51/"	"1.53"	"1.58"	"1.59"	"1.6/"
##	[373]	"1.62"	"1.64"	"1.65"	"1.67"	"1.69"	"1.7"
##	[379]	"1.72"	"1.75"	"1.75/"	"1.8"	"1.81"	"1.81/"
##	[385]	"1.82"	"1.847"	"1.86"	"1.93"	"1.99"	"1/"
##	[391]	"10"	"10.0/"	"10.3"	"10.37"	"10.5/"	"10.62"
##	[397]	"10/"	"100"	"100/"	"1000"	"1000/"	"1023"
##	[403]	"103.705"	"1034.8"	"104"	"104.1/"	"104/"	"105"
##	[409]	"105/"	"1056.7"	"1075.7"	"108.27"	"109.579"	"11.3"
##	[415]	"11.7"	"110.3"	"1104"	"112"	"112.5"	"115/"
##	[421]	"12"	"12.0/"	"12.02"	"12.17"	"12.429"	"12.5"
##	[427]	"12.5/"	"12.50/"	"12.57"	"12.8"	"12/"	"120"
##	[433]	"120.65"	"120/"	"1200/"	"122.4"	"124.03"	"125"
##	[439]	"125/"	"1267.9"	"127"	"13"	"13.3"	"13.3/"
##	[445]	"13.35"	"13.45"	"13.66"	"13.75"	"13.8"	"13.9"
##	[451]	"13.9/"	"132"	"133"	"134.80/"	"13407"	"136"
##	[457]	"136.25/"	"136/"	"137/"	"138.21"	"138.84"	"14"
##	[463]	"14.34"	"14.53"	"14.6"	"14.785"	"14/"	"1400"
##	[469]	"144"	"144.0/"	"144/"	"145.3"	"148.3"	"15"
##	[475]	"15.09"	"15.1"	"15.3"	"15.63"	"15/"	"150"
##	[481]	"150.46"	"150.5"	"150/"	"1500"	"1507"	"156"

##	[487]	"16"	"16.2"	"16.4"	"16.69"	"16/"	"167.9"
##	[493]	"168/"	"17"	"17.24"	"17.32"	"17.5"	"17.6"
##	[499]	"17.92"	"170"	"170.52"	"173.088"	"176.5"	"18"
##	[505]	"18.5"	"180"	"183.4"	"187.208"	"19.24"	"19.7"
##	[511]	"190.2"	"191.044"	"193.59"	"2"	"2.0/"	"2.008"
##	[517]	"2.1"	"2.15"	"2.16"	"2.17"	"2.18/"	"2.19"
##	[523]	"2.2"	"2.2/"	"2.217"	"2.24/"	"2.26"	"2.27/"
##	[529]	"2.3"	"2.31"	"2.38"	"2.4"	"2.403"	"2.403/"
##	[535]	"2.44"	"2.46"	"2.462"	"2.48"	"2.5"	"2.5/"
##	[541]	"2.556"	"2.6"	"2.6/"	"2.60/"	"2.608"	"2.61"
##	[547]	"2.63"	"2.644"	"2.667"	"2.68"	"2.689"	"2.75"
##	[553]	"2.761"	"2.78"	"2.79"	"2.8"	"2.82"	"2.84"
##	[559]	"2.844"	"2.85"	"2.86"	"2.89"	"2.9"	"2.936"
##	[565]	"2.94"	"2.947"	"2.96"	"2.967"	"2.99"	"2/"
##	[571]	"20"	"20.0/"	"20.3"	"20.4"	"20.6"	"20.636"
##	[577]	"20/"	"200"	"200/"	"203.6"	"2051"	"208.9"
##	[583]	"21.06"	"21.19"	"21.209"	"21.4"	"21.418"	"21.5"
##	[589]	"21.6"	"21/"	"212"	"217"	"218.89"	"22"
##	[595]	"22.2"	"22.4"	"22.59"	"222.65"	"224.05/"	"2240.5"
##	[601]	"225"	"228.8"	"23.37"	"23.37/"	"23.54"	"23.6"
##	[607]	"23.8"	"23.93"	"232.37"	"239.1"	"24"	"24.33"
##	[613]	"24.46"	"24.54"	"24.96"	"240/"	"241.3"	"242"
##	[619]	"242.45"	"245.5"	"249.23/"	"25"	"25.0/"	"25.39"
##	[625]	"25/"	"250"	"250/"	"251.3"	"251.3/"	"2558"
##	[631]	"26.025/"	"26.63"	"26.9"	"26/"	"263.44"	"2675"
##	[637]	"268"	"27.16"	"27.2"	"272.5/"	"276"	"28"
##	[643]	"28.2"	"28.4"	"28.81"	"28/"	"286/"	"288"
##	[649]	"29.79"	"296.62"	"3"	"3.0/"	"3.018"	"3.03"
##	[655]	"3.09"	"3.11"	"3.12"	"3.151"	"3.18"	"3.2"
##	[661]	"3.21"	"3.23"	"3.28"	"3.3"	"3.31"	"3.313"
##	[667]	"3.336"	"3.36"	"3.38"	"3.392"	"3.392/"	"3.4"
##	[673]	"3.41"	"3.42"	"3.4859"	"3.49"	"3.5"	"3.5/"
##	[679]	"3.53"	"3.56"	"3.59"	"3.6"	"3.62"	"3.68"
##	[685]	"3.7"	"3.743"	"3.747"	"3.75"	"3.75/"	"3.789"
##	[691]	"3.8"	"3.8/"	"3.82"	"3.885"	"3.9"	"3/"
##	[697]	"30"	"30.3"	"30.6"	"30.7"	"30/"	"300"
##	[703]	"31.16"	"31.3"	"31.37"	"31.42"	"311.9"	"32"
##	[709]	"32.4"	"32.8"	"32/"	"326.69"	"33.6"	"3390"
##	[715]	"34.3"	"34.37"	"34.96"	"35"	"35.183"	"35.813"
##	[721]	"350/"	"36/"	"360/"	"364.07"	"365.6"	"37"
##	[727]	"37.01"	"37.5/"	"37.83"	"376.3"	"382.31"	"383"
##	[733]	"39"	"39.32"	"39.37"	"4"	"4.0/"	"4.07"
##	[739]	"4.08"	"4.1"	"4.13"	"4.15"	"4.17"	"4.24"
##	[745]	"4.27"	"4.28"	"4.29"	"4.316"	"4.32"	"4.34"
##	[751]	"4.369"	"4.37"	"4.375/"	"4.383"	"4.39/"	"4.4"
##	[757]	"4.411"	"4.44"	"4.45"	"4.485"	"4.507"	"4.51"
##	[763]	"4.514"	"4.52"	"4.53"	"4.546"	"4.552"	"4.6"
##	[769]	"4.61"	"4.627"	"4.64"	"4.66"	"4.671"	"4.679"
##	[775]	"4.71"	"4.75"	"4.76"	"4.8"	"4.82"	"4.83"
##	[781]	"4.92"	"4.93"	"4/"	"40"	"40.0/"	"40.019"
##	[787]	"40.44"	"40.7"	"40/"	"400/"	"41.6"	"41.94"
##	[793]	"42"	"42.9"	"42.9/"	"43.02"	"43.4"	"43.59"
##	[799]	"43.7"	"4399"	"440"	"447.82"	"45.37"	"45.4/"
##	[805]	"45.9"	"450"	"451"	"46.7"	"46.763"	"46/"

```
## [811] "461.2"      "462.9"      "47"         "48"         "48.2"       "48.96"
## [817] "48/"        "480"        "49.42"      "5"          "5.0/"       "5.01"
## [823] "5.06"       "5.07"       "5.08"       "5.1"        "5.1/"       "5.101"
## [829] "5.11"       "5.16"       "5.18"       "5.2"        "5.25"       "5.27"
## [835] "5.28"       "5.3"        "5.3/"       "5.35"       "5.36"       "5.37"
## [841] "5.38"       "5.38/"      "5.4"        "5.46"       "5.49"       "5.53"
## [847] "5.56"       "5.6"        "5.64"       "5.7"        "5.71"       "5.73"
## [853] "5.77"       "5.79"       "5.8"        "5/"         "50"         "50.28"
## [859] "50/"        "500"        "500/"       "503.6"      "51.16"      "51.32"
## [865] "52.05/"     "52.2"       "52.6/"      "53"         "537"        "54.67"
## [871] "545/"       "56"         "56.35"      "56.73"      "57"         "57.7"
## [877] "5737"       "59"         "59.36"      "6"          "6.0/"       "6.03"
## [883] "6.13"       "6.135"      "6.25"       "6.27"       "6.36"       "6.5"
## [889] "6.55"       "6.57"       "6.6"        "6.7"        "6.7/"       "6.76"
## [895] "6.80/"      "6.83"       "6.95"       "6/"         "60"         "60.1"
## [901] "60.7"       "60/"        "600"        "61"         "62.5/"      "620.21"
## [907] "63.54"      "635.4"      "636.5"      "64.6"       "65.14"      "65.37"
## [913] "65.56"      "65.68"      "675.4"      "683.2"      "6830"       "695.5"
## [919] "7"          "7.07"       "7.14"       "7.32"       "7.379"      "7.379/"
## [925] "7.5"        "7.5/"       "7.6"        "7.7"        "7.72"       "7.8"
## [931] "7/"         "70"         "70/"        "71"         "71.3"       "71/"
## [937] "721.6"      "74.631"     "74.9"       "75"         "75.26"      "75.27"
## [943] "75/"        "750"        "754.2"      "76"         "76.035"     "76/"
## [949] "7839"       "788.5"      "8"          "8.0/"       "8.03"       "8.1"
## [955] "8.38"       "8.51"       "8.6"        "8.86"       "8/"         "80"
## [961] "80.66"      "80.9"       "80/"        "800"        "8000"       "80000"
## [967] "801"        "81.09"      "83"         "83.97"      "8352"       "84"
## [973] "84/"        "85.3"       "85.47"      "85.8"       "85.8/"      "86.9"
## [979] "87.1"       "87.5"       "9"          "9.07"       "9.49"       "9.5"
## [985] "9.5858"     "9.6"        "9.89"       "9.94"       "90"         "90/"
## [991] "900"        "93.21"      "94"         "95.48"      "95.8"       "96"
## [997] "96/"        "97.425"     "98.43"      "98.43/"     "99.063"     "99.75"
## [1003] "99.82"      "995"        "NR"         "NR/"
```

```
class(ecotoxdf$Conc.1..Author.)
```

```
## [1] "factor"
```

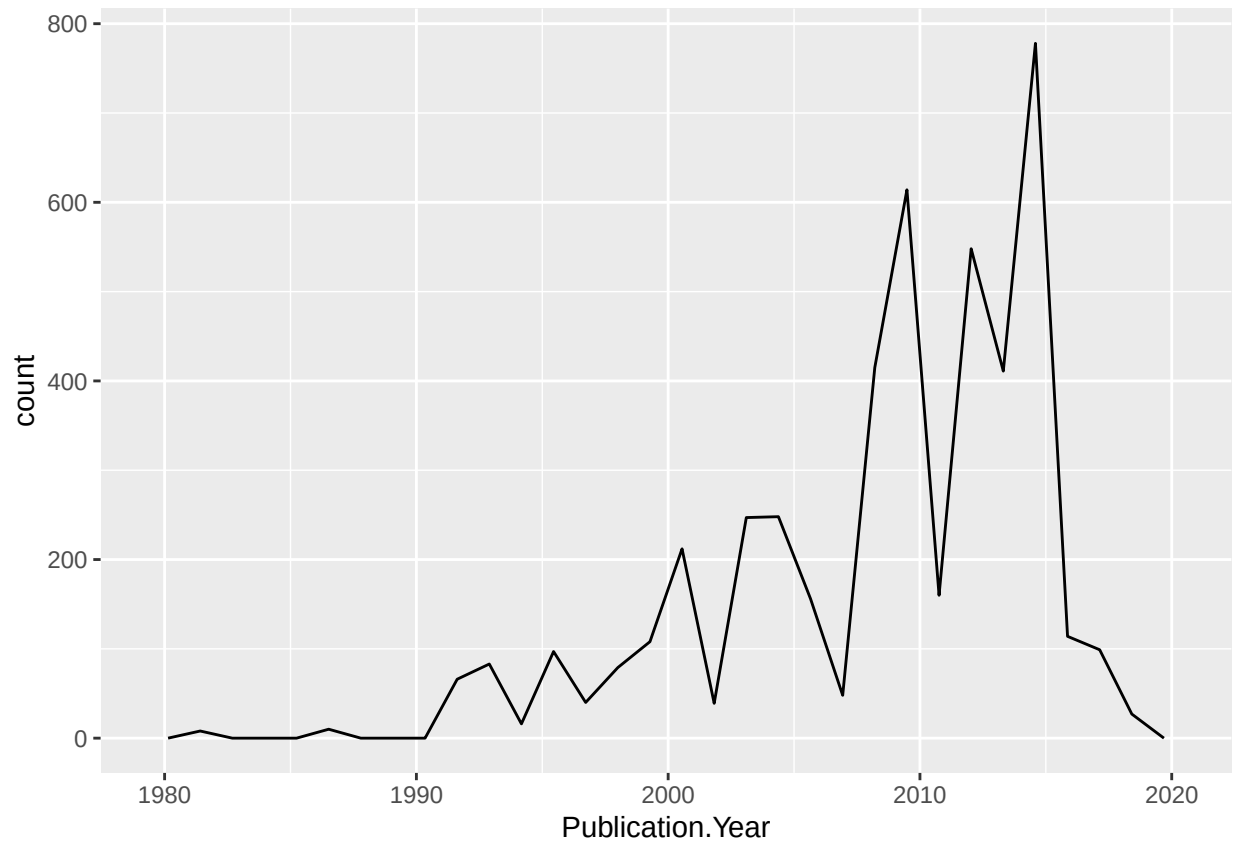
Answer: The class is currently a factor - this is because there are an apparent variety of conventions in the data. There are a mix of integers and floating points, there are slashes appended to the end of many entries (seemingly to indicate “per” the units found in the next column), and there are even inequalities and approximations of the data.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(ecotoxdf) +
  geom_freqpoly(aes(x = Publication.Year))
```

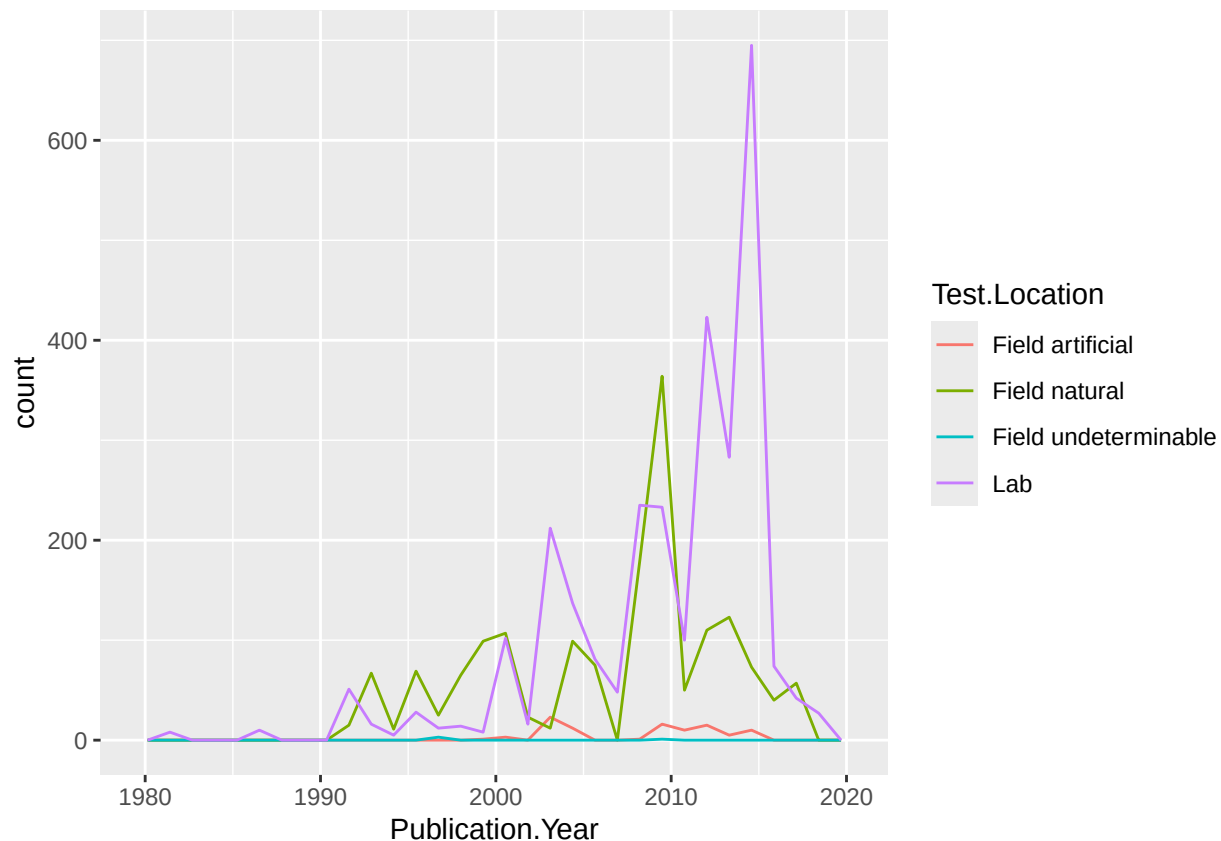
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(ecotoxdf) +  
  geom_freqpoly(aes(x = Publication.Year, colour = Test.Location))
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



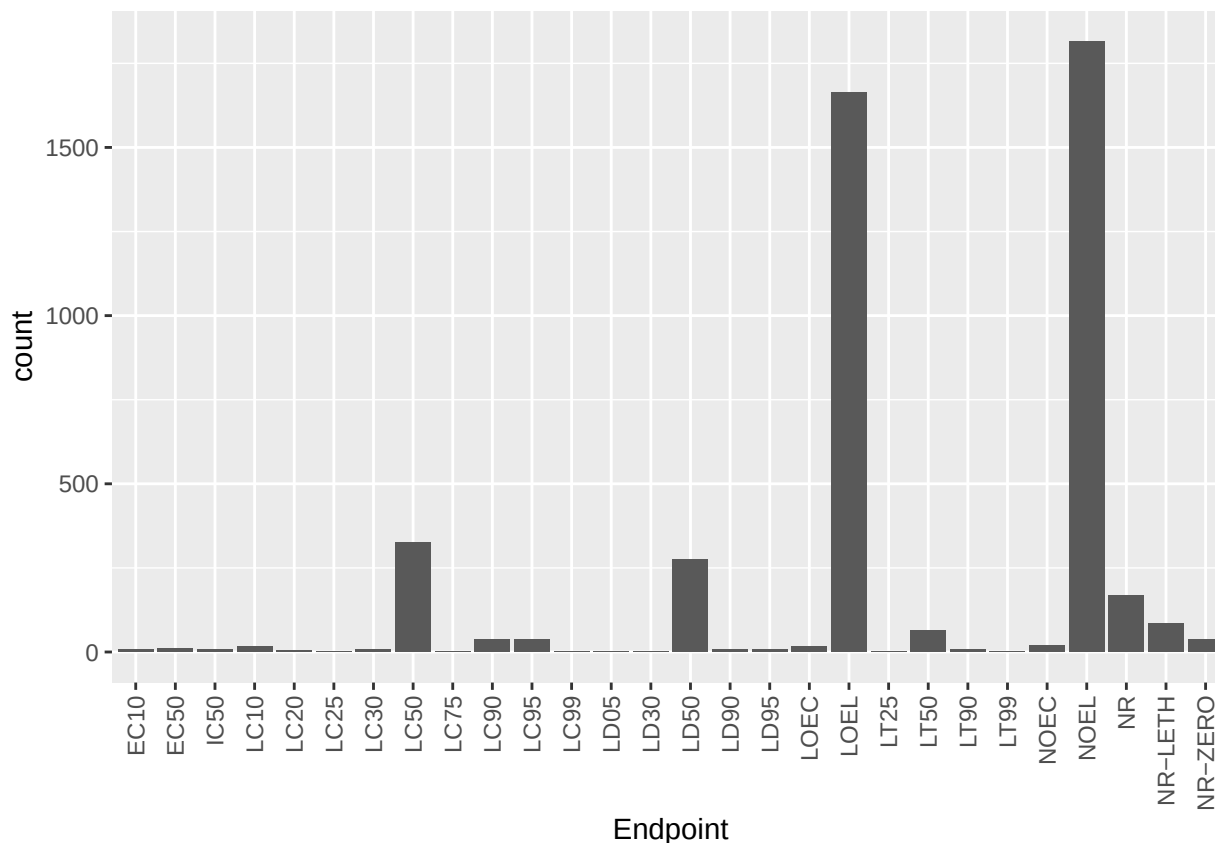
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The two most common test locations are Lab & “Field natural”; field natural is the predominant source from the early 1990s to 2000, when Lab becomes the predominant source for most of the remainder of the period.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(ecotoxdf)+
  geom_bar(aes(x = Endpoint))+
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: NOEL No-observable-effect-level: highest dose (concentration) producing effects not significantly different from responses of controls according to author's reported statistical test (NOEAL/NOEC) & LOEL Lowest-observable-effect-level: lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEAL/LOEC)

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
library(lubridate)
class(neondf$collectDate)
```

```
## [1] "factor"
```

```
neondf$collectDate <- ymd(neondf$collectDate)
class(neondf$collectDate)
```

```
## [1] "Date"
```

```
unique(neondf$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(neondf$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

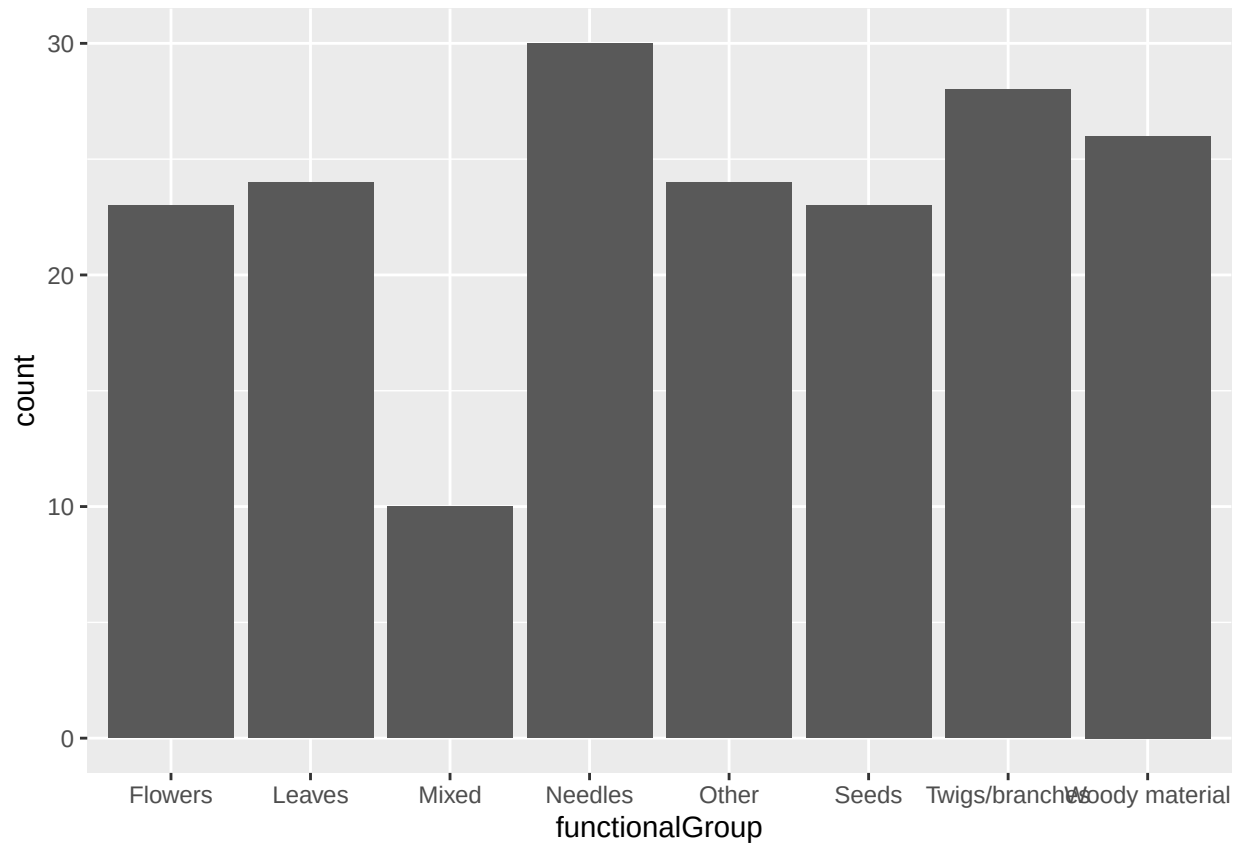
```
summary(neondf$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14       8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: Unique provides only the list of unique values, while summary also provides counts of those unique values in the dataset

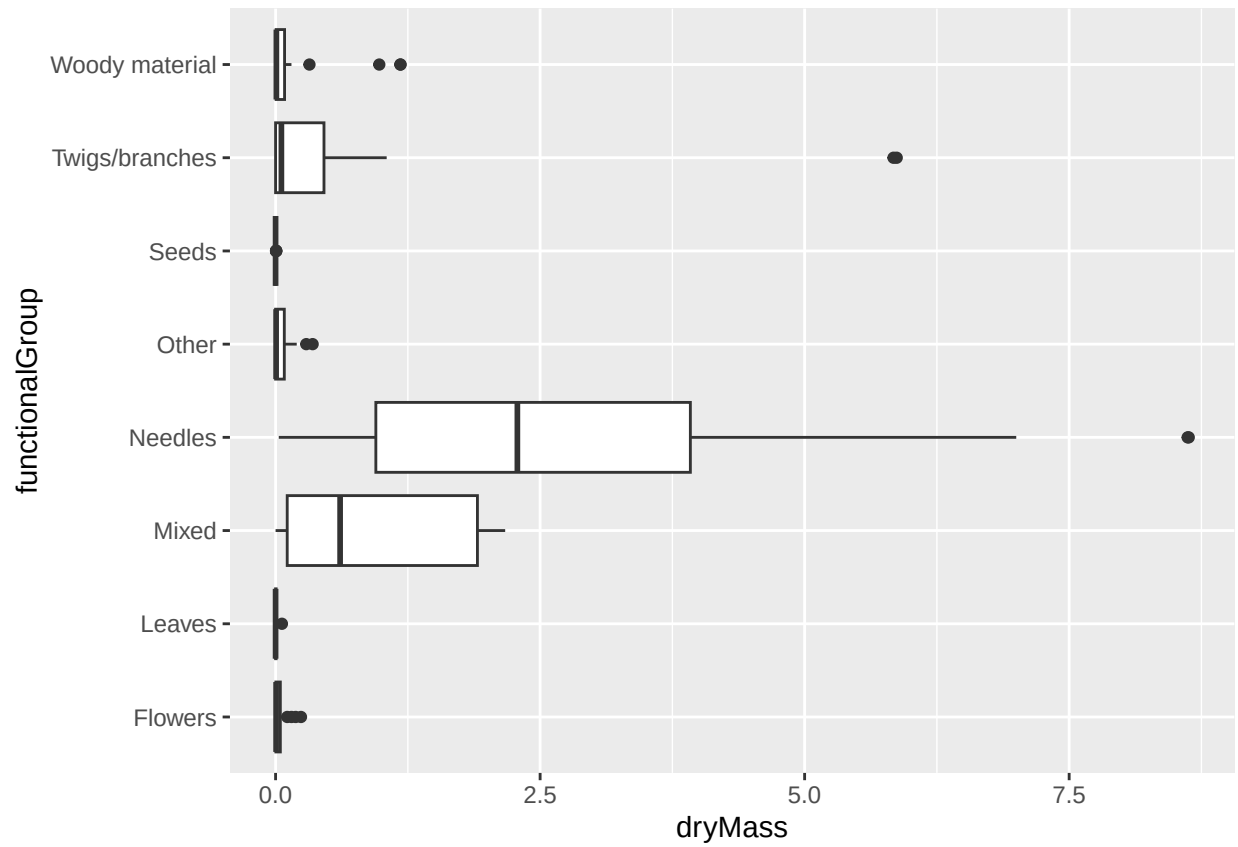
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(neondf)+  
  geom_bar(aes(x=functionalGroup))
```

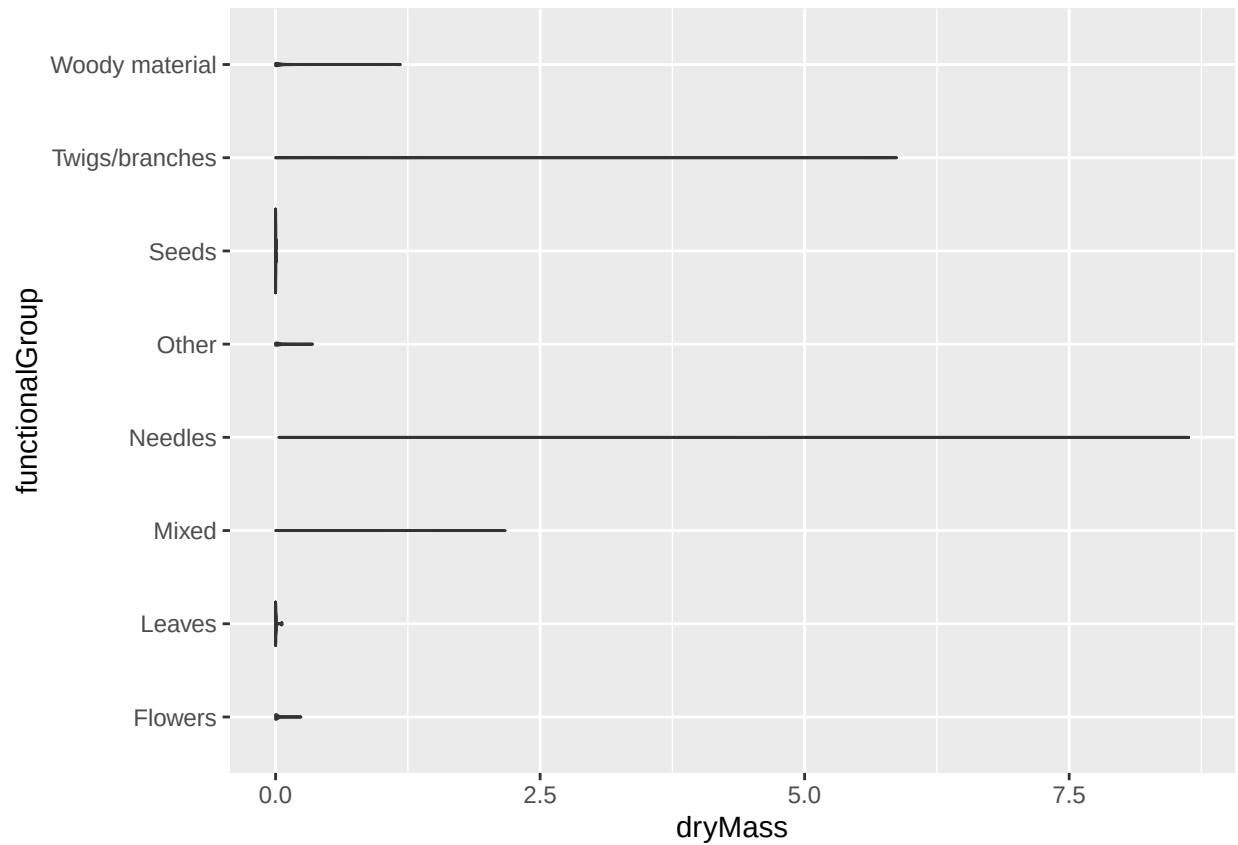


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(neondf) +  
  geom_boxplot(aes(x=dryMass, y=functionalGroup))
```



```
ggplot(neondf)+  
  geom_violin(aes(x=dryMass, y=functionalGroup),  
    draw_quantiles = c(0.25, 0.5, 0.75))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: the boxplot is more effective because there don't seem to be enough observations in the dataset for the violin plot to adjust shape to frequency.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Seeds, leaves and flowers